Method of Characteristics (MOC) Governing Equations and Formulation

1 Introduction

The Method of Characteristics (MOC) is a classical technique used to solve hyperbolic partial differential equations. In compressible flow, it is especially important for two-dimensional, steady, inviscid, irrotational, and isentropic supersonic flows. The main reason for its importance is that, in the supersonic regime, information propagates along specific directions in the flow field. These directions are called *characteristic lines*.

Instead of solving the governing partial differential equation everywhere at once, the MOC transforms the problem into relations that hold along these characteristic lines. This makes it possible to construct the flow field point by point and, in nozzle design problems, to generate wall contours that produce the desired expansion behavior.

In simple terms, the Method of Characteristics tells us three things:

1. what the governing equation of the flow is,
2. along which directions the flow information travels,
3. which quantities remain related along those directions.

Once these are known, the flow can be built geometrically and algebraically.

2 Assumptions

The standard two-dimensional MOC formulation used in elementary supersonic nozzle design is based on the following assumptions:

$$\text{steady flow,} \tag{1}$$
$$\text{inviscid flow,} \tag{2}$$
$$\text{compressible flow,} \tag{3}$$
$$\text{irrotational flow,} \tag{4}$$
$$\text{isentropic flow.} \tag{5}$$

These assumptions are important because they reduce the full compressible flow problem to a form that can be treated analytically.

3 Velocity Components and Basic Definitions

For a two-dimensional flow, the velocity vector is written as

$$V = u\,\hat{\imath} + v\,\hat{\jmath}, \tag{6}$$

where u and v are the velocity components in the x and y directions, respectively.

The magnitude of the velocity is

$$V = \sqrt{u^2 + v^2}. \tag{7}$$

The local flow angle θ is defined by

$$\tan\theta = \frac{v}{u}. \tag{8}$$

This angle describes the local direction of the velocity vector relative to the x-axis.

4 Continuity Equation

The continuity equation for steady compressible flow is

$$\nabla \cdot (\rho V) = 0. \tag{9}$$

In Cartesian coordinates, this becomes

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0. \tag{10}$$

Expanding the derivatives gives

$$\rho\frac{\partial u}{\partial x} + \rho\frac{\partial v}{\partial y} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} = 0. \tag{11}$$

Dividing by ρ:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + u\frac{1}{\rho}\frac{\partial \rho}{\partial x} + v\frac{1}{\rho}\frac{\partial \rho}{\partial y} = 0. \tag{12}$$

Since

$$\frac{1}{\rho}\frac{\partial \rho}{\partial x} = \frac{\partial \ln\rho}{\partial x}, \qquad \frac{1}{\rho}\frac{\partial \rho}{\partial y} = \frac{\partial \ln\rho}{\partial y}, \tag{13}$$

the continuity equation may be written as

$$u_x + v_y + u(\ln\rho)_x + v(\ln\rho)_y = 0. \tag{14}$$

This form is useful because it connects velocity derivatives with density variations.

5 Energy Equation and Isentropic Relation

For steady, adiabatic, inviscid flow, the stagnation enthalpy is constant:

$$H = C_pT + \frac{V^2}{2} = \text{constant}. \tag{15}$$

Under isentropic conditions, the standard temperature relation is

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2, \tag{16}$$

where

$$M = \frac{V}{a} \tag{17}$$

is the Mach number and a is the local speed of sound.

The speed of sound is

$$a^2 = \gamma RT. \tag{18}$$

These relations are essential because they connect the thermodynamic state to the velocity field.

6 Irrotational Flow and Velocity Potential

For irrotational flow,

$$\nabla \times V = 0. \tag{19}$$

In two dimensions,

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0. \tag{20}$$

Because the flow is irrotational, a velocity potential ϕ can be introduced such that

$$u = \frac{\partial \phi}{\partial x}, \qquad v = \frac{\partial \phi}{\partial y}. \tag{21}$$

This is useful because it reduces the unknowns to a single scalar function ϕ.

7 The Governing Potential Equation

When the continuity equation, isentropic relation, and potential-flow assumption are combined, the governing equation for two-dimensional compressible potential flow becomes

$$(a^2 - u^2)\phi_{xx} - 2uv\phi_{xy} + (a^2 - v^2)\phi_{yy} = 0. \tag{22}$$

This is the central partial differential equation in the MOC formulation.

An equivalent velocity form is

$$\left(1 - \frac{u^2}{a^2}\right) u_x + \left(1 - \frac{v^2}{a^2}\right) v_y - \frac{uv}{a^2}(u_y + v_x) = 0. \tag{23}$$

Both forms describe the same physics. The potential form is usually more convenient when deriving the characteristic directions.

8 Why the Type of PDE Matters

A second-order PDE can be elliptic, parabolic, or hyperbolic. The behavior of the solution changes completely depending on its type.

For a general second-order equation

$$A\phi_{xx} + 2B\phi_{xy} + C\phi_{yy} = 0, \tag{24}$$

the classification is determined by

$$B^2 - AC. \tag{25}$$

For the compressible potential equation,

$$A = a^2 - u^2, \qquad B = -uv, \qquad C = a^2 - v^2. \tag{26}$$

Therefore,

$$B^2 - AC = u^2v^2 - (a^2 - u^2)(a^2 - v^2) \tag{27}$$

$$= a^2(u^2 + v^2 - a^2). \tag{28}$$

Since $u^2 + v^2 = V^2$, this becomes

$$B^2 - AC = a^2(V^2 - a^2) = a^4(M^2 - 1). \tag{29}$$

Hence:

$$M > 1 \Rightarrow \text{hyperbolic}, \tag{30}$$

$$M = 1 \Rightarrow \text{parabolic}, \tag{31}$$

$$M < 1 \Rightarrow \text{elliptic}. \tag{32}$$

This is the key reason MOC is used for **supersonic** flow. In the supersonic region, the equation is hyperbolic, and hyperbolic equations possess characteristic lines.

9 Characteristic Equation

To find the characteristic directions, start from the general PDE

$$A\phi_{xx} + 2B\phi_{xy} + C\phi_{yy} = 0. \tag{33}$$

The characteristic equation is

$$A(dy)^2 - 2B\,dx\,dy + C(dx)^2 = 0. \tag{34}$$

Substituting

$$A = a^2 - u^2, \qquad B = -uv, \qquad C = a^2 - v^2, \tag{35}$$

we obtain

$$(a^2 - u^2)(dy)^2 + 2uv\,dx\,dy + (a^2 - v^2)(dx)^2 = 0. \tag{36}$$

Dividing by dx^2 gives

$$(a^2 - u^2)\left(\frac{dy}{dx}\right)^2 + 2uv\left(\frac{dy}{dx}\right) + (a^2 - v^2) = 0. \tag{37}$$

This is a quadratic equation for the slope yx.

10 Characteristic Slopes

Solving the quadratic equation gives

$$\frac{dy}{dx} = \frac{-uv \pm \sqrt{u^2v^2 - (a^2 - u^2)(a^2 - v^2)}}{a^2 - u^2}. \tag{38}$$

This algebraic expression is correct but not yet physically transparent. We now rewrite it in terms of the flow angle θ and the Mach angle μ.

11 Mach Angle

For supersonic flow, the Mach angle μ is defined by

$$\sin\mu = \frac{1}{M}. \tag{39}$$

This angle describes the direction at which disturbances propagate relative to the local flow direction.

The remarkable result of the derivation is that the characteristic slopes can be written as

$$\boxed{\frac{dy}{dx} = \tan(\theta \pm \mu)}. \tag{40}$$

Thus, in a supersonic flow field, there are two characteristic directions through each point:

$$C^+ : \frac{dy}{dx} = \tan(\theta + \mu), \tag{41}$$

$$C^- : \frac{dy}{dx} = \tan(\theta - \mu). \tag{42}$$

These are the two characteristic lines.

12 Physical Meaning of the Characteristic Lines

The characteristic lines are not arbitrary mathematical curves. They represent the directions along which information propagates through the supersonic flow field.

Since the flow is supersonic, disturbances cannot travel upstream in all directions. Instead, they are confined to directions forming the Mach cone in three dimensions, or Mach lines in two dimensions. In two-dimensional MOC, these directions appear as the two characteristic lines:

$$\theta + \mu, \qquad \theta - \mu. \tag{43}$$

This is why MOC is geometrically powerful: once the local flow direction and Mach number are known, the local characteristic directions are also known.

13 Compatibility Relations

Finding the characteristic directions alone is not enough. We also need relations that hold *along* these lines.

For steady, two-dimensional, irrotational, isentropic supersonic flow, the differential compatibility relation is

$$d\theta = \pm\sqrt{M^2 - 1}\,\frac{dV}{V}. \tag{44}$$

This equation tells us how the flow angle changes as the speed changes along a characteristic direction.

Integrating between two states:

$$\theta_2 - \theta_1 = \pm \int_1^2 \sqrt{M^2 - 1}\,\frac{dV}{V}. \tag{45}$$

To simplify the notation, the integral is collected into a special function called the Prandtl–Meyer function.

14 Prandtl–Meyer Function

The Prandtl–Meyer function $\nu(M)$ is defined such that

$$d\nu = \sqrt{M^2 - 1}\,\frac{dV}{V}. \tag{46}$$

Hence,

$$\nu(M_2) - \nu(M_1) = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\,\frac{dV}{V}. \tag{47}$$

Therefore the compatibility relations become

$$\theta \pm \nu(M) = \text{constant}. \tag{48}$$

More explicitly,

$$C^+ : \theta + \nu = \text{constant}, \tag{49}$$

$$C^- : \theta - \nu = \text{constant}. \tag{50}$$

These are the most important working equations in the classical two-dimensional MOC.

15 Why These Relations Are So Important

At this point, the structure of the method becomes clear.

At any point in the supersonic flow field:

1. the two characteristic directions are known from

$$\frac{dy}{dx} = \tan(\theta \pm \mu), \tag{51}$$

2. the quantities $\theta + \nu$ and $\theta - \nu$ remain constant along the corresponding characteristic lines.

This means that if two characteristic lines intersect, the flow properties at the intersection can be determined from the known invariants carried by those lines.

That is the core idea of point-by-point construction in the Method of Characteristics.

16 The Closed-Form Prandtl–Meyer Formula

For a calorically perfect gas, the Prandtl–Meyer function is usually written as

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1}\left(\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)}\right) - \tan^{-1}\left(\sqrt{M^2-1}\right). \tag{52}$$

This function is zero at $M = 1$ and increases as the Mach number increases. Physically, it measures the amount of isentropic turning associated with a supersonic expansion.

17 Main Summary of the Theory

The complete logic of the elementary two-dimensional MOC may be summarized as follows:

1. Start with steady, inviscid, compressible, irrotational, isentropic flow.

2. Write the compressible potential equation:

$$(a^2 - u^2)\phi_{xx} - 2uv\phi_{xy} + (a^2 - v^2)\phi_{yy} = 0. \tag{53}$$

3. Show that the equation is hyperbolic when $M > 1$.

4. Derive the characteristic directions:

$$\frac{dy}{dx} = \tan(\theta \pm \mu). \tag{54}$$

5. Derive the compatibility relations:

$$\theta + \nu = \text{constant on } C^+, \qquad \theta - \nu = \text{constant on } C^-. \tag{55}$$

6. Use these relations to compute unknown flow properties at intersections of characteristic lines.

18 What Comes Next After This Theory

Once the theoretical foundation is understood, the next practical step is to use these equations in **unit processes**. In classical MOC nozzle design, the most common are:

1. interior-point process,
2. wall-point process,
3. centerline process.

These processes use the characteristic relations to determine the coordinates and flow properties of new points from previously known points. By repeating them systematically, the entire supersonic flow field can be constructed.

19 Conclusion

The Method of Characteristics is fundamentally a way of converting a hyperbolic partial differential equation into a family of ordinary relations along special curves. For two-dimensional supersonic flow, those curves are the characteristic lines whose slopes are determined by the flow angle and Mach angle:

$$\frac{dy}{dx} = \tan(\theta \pm \mu). \tag{56}$$

Along those lines, the combinations

$$\theta + \nu, \qquad \theta - \nu \tag{57}$$

remain constant. This pair of geometric and algebraic relations forms the basis of classical supersonic nozzle design and wave interaction analysis.

20 Introduction

After deriving the characteristic relations for two-dimensional, steady, inviscid, irrotational, compressible, and isentropic supersonic flow, the next step in the Method of Characteristics (MOC) is to use these relations to construct the flow field point by point.

The purpose of this stage is not merely to study the differential equations, but to generate a computational or geometric mesh in which every new point is found from previously known points. In classical MOC, this is done through three basic unit processes:

1. interior-point process,
2. centerline-point process,
3. wall-point process.

21 Characteristic Invariants

For supersonic flow, the compatibility relations are

$$K^+ = \theta + \nu = \text{constant along } C^+, \tag{58}$$

$$K^- = \theta - \nu = \text{constant along } C^-. \tag{59}$$

The characteristic directions are

$$\frac{dy}{dx} = \tan(\theta + \mu) \qquad \text{for } C^+, \tag{60}$$

$$\frac{dy}{dx} = \tan(\theta - \mu) \qquad \text{for } C^-, \tag{61}$$

where

$$\mu = \sin^{-1}\left(\frac{1}{M}\right) \tag{62}$$

is the Mach angle.

At each grid point, the main quantities are usually

$$(x,\ y,\ \theta,\ \nu,\ M,\ \mu). \tag{63}$$

22 Interior-Point Process

Consider two known points, A and B, from which two characteristic lines intersect at a new unknown point P.

Suppose:

- the C^+ characteristic comes from point A,
- the C^- characteristic comes from point B.

Then the invariants carried to point P are

$$\theta_P + \nu_P = \theta_A + \nu_A, \tag{64}$$

$$\theta_P - \nu_P = \theta_B - \nu_B. \tag{65}$$

Define

$$K_A^+ = \theta_A + \nu_A, \qquad K_B^- = \theta_B - \nu_B. \tag{66}$$

Then

$$\boxed{\theta_P = \frac{K_A^+ + K_B^-}{2}} \tag{67}$$

and

$$\boxed{\nu_P = \frac{K_A^+ - K_B^-}{2}}. \tag{68}$$

Once ν_P is known, the Mach number at point P is found from the inverse Prandtl–Meyer relation:

$$M_P = \nu^{-1}(\nu_P). \tag{69}$$

Then the Mach angle becomes

$$\mu_P = \sin^{-1}\left(\frac{1}{M_P}\right). \tag{70}$$

22.1 Coordinates of the Interior Point

To find the coordinates of point P, the two characteristic lines are approximated as straight segments between the known and unknown points.

The slope of the characteristic coming from point A is approximated by

$$m_A = \tan\left[\frac{\theta_A + \theta_P}{2} + \frac{\mu_A + \mu_P}{2}\right]. \tag{71}$$

The slope of the characteristic coming from point B is approximated by

$$m_B = \tan\left[\frac{\theta_B + \theta_P}{2} - \frac{\mu_B + \mu_P}{2}\right]. \tag{72}$$

The equations of the two lines are

$$y - y_A = m_A(x - x_A), \tag{73}$$

$$y - y_B = m_B(x - x_B). \tag{74}$$

Their intersection gives the coordinates of point P:

$$\boxed{x_P = \frac{(y_B - y_A) + m_A x_A - m_B x_B}{m_A - m_B}} \tag{75}$$

and

$$\boxed{y_P = y_A + m_A(x_P - x_A)}. \tag{76}$$

23 Centerline-Point Process

For a symmetric nozzle, the centerline is

$$y = 0. \tag{77}$$

Because of symmetry, the flow angle on the centerline must be

$$\theta = 0. \tag{78}$$

Suppose a characteristic from a known interior point A intersects the centerline at a new point P.

At point P,

$$y_P = 0, \qquad \theta_P = 0. \tag{79}$$

Using the appropriate characteristic invariant:

$$\theta_P \pm \nu_P = \theta_A \pm \nu_A. \tag{80}$$

Since $\theta_P = 0$,

$$\nu_P = \theta_A + \nu_A \qquad \text{or} \qquad \nu_P = \nu_A - \theta_A, \tag{81}$$

depending on whether the incoming characteristic is C^- or C^+.

After obtaining ν_P,

$$M_P = \nu^{-1}(\nu_P), \tag{82}$$

$$\mu_P = \sin^{-1}\left(\frac{1}{M_P}\right). \tag{83}$$

23.1 Coordinates of the Centerline Point

The slope of the incoming characteristic is approximated by

$$m_A = \tan\left[\frac{\theta_A + \theta_P}{2} \pm \frac{\mu_A + \mu_P}{2}\right]. \tag{84}$$

Since $y_P = 0$, the characteristic line equation gives

$$0 - y_A = m_A(x_P - x_A). \tag{85}$$

Therefore,

$$\boxed{x_P = x_A - \frac{y_A}{m_A}}. \tag{86}$$

24 Wall-Point Process

In nozzle design, a wall point is a point that lies on the nozzle contour. At the wall, the flow must be tangent to the surface. Therefore, if the local wall angle is θ_w, then

$$\theta_P = \theta_w. \tag{87}$$

Suppose a characteristic from a known interior point A intersects the wall at a new point P.

Using the appropriate invariant,

$$\theta_P \pm \nu_P = \theta_A \pm \nu_A. \tag{88}$$

Since θ_P is known from the wall geometry, ν_P can be found directly:

$$\nu_P = K - \theta_P \qquad \text{or} \qquad \nu_P = \theta_P - K, \tag{89}$$

depending on the characteristic family.

Then

$$M_P = \nu^{-1}(\nu_P), \tag{90}$$

$$\mu_P = \sin^{-1}\left(\frac{1}{M_P}\right). \tag{91}$$

24.1 Coordinates of the Wall Point

The characteristic slope from point A to point P is approximated as

$$m_c = \tan\left[\frac{\theta_A + \theta_P}{2} \pm \frac{\mu_A + \mu_P}{2}\right]. \tag{92}$$

The wall slope is approximated by

$$m_w = \tan\left(\frac{\theta_A + \theta_P}{2}\right) \tag{93}$$

or, in simpler formulations,

$$m_w = \tan\theta_P. \tag{94}$$

If the previous wall point is denoted by W, then the wall line is

$$y - y_W = m_w(x - x_W), \tag{95}$$

and the characteristic line is

$$y - y_A = m_c(x - x_A). \tag{96}$$

Their intersection gives the new wall-point coordinates.

25 Overall Grid Construction Logic

The full MOC mesh is constructed iteratively. The procedure is:

1. start with an initial data line or a set of known points,
2. use two intersecting characteristics to compute interior points,
3. use symmetry to compute centerline points,
4. use wall tangency to compute wall points,
5. repeat the process until the full flow region is covered.

Thus, the entire solution is built from previously known points. Every newly found point becomes a source for future characteristics.

26 Practical Computational Sequence

For each new point, the usual sequence is:

1. identify which characteristic relations apply,
2. write the appropriate invariant equations,
3. solve for θ and ν,
4. compute M from the inverse Prandtl–Meyer function,
5. compute $\mu = \sin^{-1}(1/M)$,
6. write the characteristic line equations,
7. solve for the coordinates (x, y).

27 Summary

The Method of Characteristics becomes practical only after learning how new points are generated. The three essential unit processes are:

$$\text{interior point: characteristic} + \text{characteristic,} \tag{97}$$

$$\text{centerline point: characteristic} + \text{symmetry,} \tag{98}$$

$$\text{wall point: characteristic} + \text{wall tangency.} \tag{99}$$

These three processes form the computational backbone of classical supersonic nozzle design.